

# Solution Outputs — Team AuraQu

*Challenge Statement §5, outputs 2 and 3. The connectivity matrices (output 1) are supplied separately as `Connectivity_Matrices.csv`; the methodological argument is Section 2 of the Concept Proposal.*

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## 1. Hit List — top five predicted allosteric residues per target

**residue\_number** is the PDB number as deposited and is the one to quote. **residue\_index** is the 0-based position in the connectivity matrix. They differ for three of the four targets, because PDB numbering does not start at zero and is not always contiguous.

**The same five, as matrix indices:**

**KRAS\_G12C is the trap:** its structure genuinely numbers from 0, so its two rows coincide. Code tested only on KRAS will silently use the wrong column on the other three.

### Read the verdicts with the list

For the three targets where a drug-bound structure exists to check against, our own validation returns **NO\_SIGNAL\_IN\_APO**: the score's confidence interval overlaps that of the best trivial baseline computed on the same structure.

Every interval includes zero. **We submit the five as required and state plainly that we cannot certify them** — the conclusion our proposal reaches about the field's published numbers, applied to our own.

1NKP (c-Myc) has no drug-bound structure in the PDB, so no ground truth exists to score against. Its five are a genuinely prospective prediction, labelled **unverified**, from a four-operator consensus.

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## 2. Known limitations, measured

**Each hit list is one draw.** Ten genuine G12C apo structures span AUC **0.408–0.595** (median 0.482); substituting the cardiac apo structure moved AUC by **0.27**. A different, equally defensible apo deposition would give a different five.

**A classical cavity detector beats our walk on one of three targets and loses on another.** Re-run fresh against the currently shipped structures: fpocket 0.860 vs our 0.541 on BCR-ABL1; 0.420 vs 0.514 on KRAS G12C; 0.535 vs 0.548 on cardiac myosin. We report this because fpocket is a stage inside our own pipeline, and

because it is evidence the benchmark is easier than it looks.

**Pooled AUC across proteins carries a protein-identity floor.** A score with all within-structure position information destroyed — every residue replaced by its own protein's mean — reaches pooled **AUC 0.65**. Every AUC in this submission is computed per structure and averaged, never pooled; we audited that rather than assuming it.

**Our own orthosteric-exclusion filter never fires on two of the three scoreable apo inputs, and this is why BCR-ABL1's #5 hit is the gate-keeper residue, not a distal one.** `func_ligand` excludes active-site contacts by exact ligand-code match against the input structure; BCR-ABL1's declared code is NIL (nilotinib), present only in the holo structure 5M04 and absent from the apo input 10PL (which carries MYR/P16 instead), and cardiac myosin's list omits V04, the ADP-vanadate transition-state mimic sitting in 8QYP beside the ADP that is listed. The lookup fails closed rather than erroring, so a stale or incomplete code produces no signal that exclusion did not happen. Named here rather than fixed: with under 36 hours to the deadline, re-running invalidates every number already in this package. TASK-0386 tracks the finding; TASK-0391 tracks the fix, scoped for after submission.

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## 3. Methodological report, per target

**KRAS\_G12C — 4LDJ**

**BCR\_ABL1 — 10PL**

**CARDIAC\_MYOSIN — 8QYP**

**MYC\_MAX — 1NKP**

Our own configuration records that this target has no folded-state allosteric pocket (`allosteric_pocket_exists: false`, `config/targets.yaml` — Myc/Max are IDPs with no surface pocket in the folded dimer); we supply the five because the challenge requires them, and we report that our best candidate site scores 0.161 druggability against our own 0.5 fpocket druggable/non-druggable threshold.

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### Provenance

Produced by the pipeline at [github.com/Oussema-t/Quantum\\_allosteric-scanner](https://github.com/Oussema-t/Quantum_allosteric-scanner) (branch bartosz), published 2026-09-15. The five

| target         | structure | #1         | #2         | #3         | #4         | #5         |
|----------------|-----------|------------|------------|------------|------------|------------|
| KRAS_G12C      | 4LDJ      | <b>31</b>  | <b>122</b> | <b>33</b>  | <b>121</b> | <b>29</b>  |
| BCR_ABL1       | 1OPL      | <b>402</b> | <b>311</b> | <b>310</b> | <b>301</b> | <b>338</b> |
| CARDIAC_MYOSIN | 8QYP      | <b>682</b> | <b>683</b> | <b>681</b> | <b>680</b> | <b>133</b> |
| MYC_MAX        | 1NKP      | <b>943</b> | <b>246</b> | <b>925</b> | <b>226</b> | <b>243</b> |

| target         | #1  | #2  | #3  | #4  | #5  |
|----------------|-----|-----|-----|-----|-----|
| KRAS_G12C      | 31  | 122 | 33  | 121 | 29  |
| BCR_ABL1       | 321 | 230 | 229 | 220 | 257 |
| CARDIAC_MYOSIN | 611 | 612 | 610 | 609 | 94  |
| MYC_MAX        | 46  | 132 | 28  | 112 | 129 |

residues in each list match the Concept Proposal's own table exactly — checked, not assumed.

| target         | AUC   | floor (trivial baseline) | paired score - floor, 95% CI | detectable<br>$\Delta$ AUC |
|----------------|-------|--------------------------|------------------------------|----------------------------|
| KRAS_G12C      | 0.514 | 0.529 (hop_from_seed)    | -0.015 [-0.100, +0.087]      | 0.093                      |
| BCR_ABL1       | 0.541 | 0.503 (hop_from_seed)    | +0.038 [-0.057, +0.130]      | 0.094                      |
| CARDIAC_MYOSIN | 0.548 | 0.454 (hop_from_seed)    | +0.095 [-0.008, +0.201]      | 0.104                      |

Target: KRAS\_G12C | Structure: 4LDJ (apo)

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#### HEADLINE VERDICT

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[NOTE] AUC values below use the decoherent/time-averaged CTQW limit (a spectral overlap between source and residue eigenvector components), not a coherent quantum-walk snapshot -- all phase information is averaged out by construction. See TASK-0097 / REVIEW-2026-07-13 finding P2-B.

```

AUC_apo_Hnew_default      : 0.514
AUC_apo_H10_baseline      : 0.346
AUC_apo_Hnew_optimised    : 0.514
AUC_holo_Hnew_optimised   : N/A
AUC_ctqw_mean             : 0.514
AUC_heat_mean             : 0.342
most_impactful_term       : V_M
least_impactful_term      : V_T
mean_rho_apo_holo         : N/A
mean_jacc20               : N/A
coherence_auc_range       : N/A
coherence_auc_at_gamma0   : N/A
coherence_classification   : N/A

_diagnosis                 : NO_SIGNAL_IN_APO
score 95% CI               : 0.514 [0.348, 0.698] (95% block-bootstrap CI)
floor 95% CI               : 0.529 [0.346, 0.716] (95% block-bootstrap CI)
CIs overlap?               : YES -- statistically indistinguishable from the floor
                           at this confidence level

```

#### DECISION-SUPPORT RECOMMENDATION

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- 1) Operator gain over H10 baseline (apo): DAUC = +0.168 (meaningful). -- but not statistically distinguishable from the trivial floor at 95% CI (see \_diagnosis above); read this point estimate as a lead, not an established gain
- 2) CTQW vs ground-state relaxation on H\_new (same operator, not a classical-diffusion comparison -- see TASK-0095): DAUC = +0.171 (CTQW genuinely helps). -- but not statistically distinguishable from the trivial floor at 95% CI (see \_diagnosis above); read this point estimate as a lead, not an established gain
- 3) Most/least impactful potential terms: most = V\_M, least = V\_T.

Target: BCR\_ABL1 | Structure: 1OPL (apo)

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## HEADLINE VERDICT

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[NOTE] AUC values below use the decoherent/time-averaged CTQW limit (a spectral overlap between source and residue eigenvector components), not a coherent quantum-walk snapshot -- all phase information is averaged out by construction. See TASK-0097 / REVIEW-2026-07-13 finding P2-B.

|                          |   |                                                       |
|--------------------------|---|-------------------------------------------------------|
| AUC_apo_Hnew_default     | : | 0.541                                                 |
| AUC_apo_H10_baseline     | : | 0.520                                                 |
| AUC_apo_Hnew_optimised   | : | 0.541                                                 |
| AUC_holo_Hnew_optimised  | : | N/A                                                   |
| AUC_ctqw_mean            | : | 0.541                                                 |
| AUC_heat_mean            | : | 0.664                                                 |
| most_impactful_term      | : | V_B                                                   |
| least_impactful_term     | : | V_C                                                   |
| mean_rho_apo_holo        | : | N/A                                                   |
| mean_jacc20              | : | N/A                                                   |
| coherence_auc_range      | : | N/A                                                   |
| coherence_auc_at_gamma0  | : | N/A                                                   |
| coherence_classification | : | N/A                                                   |
|                          |   |                                                       |
| _diagnosis               | : | NO_SIGNAL_IN_APO                                      |
| score 95% CI             | : | 0.541 [0.375, 0.689] (95% block-bootstrap CI)         |
| floor 95% CI             | : | 0.503 [0.350, 0.638] (95% block-bootstrap CI)         |
| CIs overlap?             | : | YES -- statistically indistinguishable from the floor |
| at this confidence level |   |                                                       |

## DECISION-SUPPORT RECOMMENDATION

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- 1) Operator gain over H10 baseline (apo): DAUC = +0.020 (marginal). -- but not statistically distinguishable from the trivial floor at 95% CI (see \_diagnosis above); read this point estimate as a lead, not an established gain
- 2) CTQW vs ground-state relaxation on H\_new (same operator, not a classical-diffusion comparison -- see TASK-0095): DAUC = -0.123 (within graph-kernel noise -- CTQW does NOT add biological information beyond the operator).
- 3) Most/least impactful potential terms: most = V\_B, least = V\_C.

Target: CARDIAC\_MYOSIN | Structure: 8QYP (apo, Bos taurus)

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## HEADLINE VERDICT

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[NOTE] AUC values below use the decoherent/time-averaged CTQW limit (a spectral overlap between source and residue eigenvector components), not a coherent quantum-walk snapshot -- all phase information is averaged out by construction. See TASK-0097 / REVIEW-2026-07-13 finding P2-B.

|                          |   |                                                       |
|--------------------------|---|-------------------------------------------------------|
| AUC_apo_Hnew_default     | : | 0.548                                                 |
| AUC_apo_H10_baseline     | : | 0.468                                                 |
| AUC_apo_Hnew_optimised   | : | 0.548                                                 |
| AUC_holo_Hnew_optimised  | : | N/A                                                   |
| AUC_ctqw_mean            | : | 0.548                                                 |
| AUC_heat_mean            | : | 0.514                                                 |
| most_impactful_term      | : | V_B                                                   |
| least_impactful_term     | : | V_R                                                   |
| mean_rho_apo_holo        | : | N/A                                                   |
| mean_jacc20              | : | N/A                                                   |
| coherence_auc_range      | : | N/A                                                   |
| coherence_auc_at_gamma0  | : | N/A                                                   |
| coherence_classification | : | N/A                                                   |
|                          |   |                                                       |
| _diagnosis               | : | NO_SIGNAL_IN_APO                                      |
| score 95% CI             | : | 0.548 [0.372, 0.774] (95% block-bootstrap CI)         |
| floor 95% CI             | : | 0.454 [0.301, 0.671] (95% block-bootstrap CI)         |
| CIs overlap?             | : | YES -- statistically indistinguishable from the floor |
| at this confidence level |   |                                                       |

## DECISION-SUPPORT RECOMMENDATION

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- 1) Operator gain over H10 baseline (apo): DAUC = +0.081 (meaningful). -- but not statistically distinguishable from the trivial floor at 95% CI (see \_diagnosis above); read this point estimate as a lead, not an established gain
- 2) CTQW vs ground-state relaxation on H\_new (same operator, not a classical-diffusion comparison -- see TASK-0095): DAUC = +0.035 (within graph-kernel noise -- CTQW does NOT add biological information beyond the operator).
- 3) Most/least impactful potential terms: most = V\_B, least = V\_R.

=== MYC\_MAX -- NO GROUND TRUTH ===

No AUC, no ceiling, and no proximity-floor check are computed or reported for this target: no holo/bound structure exists for this allosteric question.

This is this target's own documented status (config/targets.yaml), not a scoring failure or an omission -- per this project's own convention, an honest NO is reported explicitly, not silently dropped.

Consensus prediction across 4 independent operators (H\_new\_default, H10\_disorder\_suppressed, H2\_combinatorial\_laplacian, H14\_anm\_pinv\_trace), top-5:

1. residue 943 -- 4/4 operators agree (top-5), mean occupancy=0.0698
2. residue 246 -- 3/4 operators agree (top-5), mean occupancy=0.0599
3. residue 925 -- 3/4 operators agree (top-5), mean occupancy=0.0494
4. residue 226 -- 2/4 operators agree (top-5), mean occupancy=0.0413
5. residue 243 -- 2/4 operators agree (top-5), mean occupancy=0.0405

Confidence: unverified -- at least one residue appears in all 4 operators' own top-5, but operator agreement is not validated confidence: no ground truth exists for this target to check against, and the Concept Proposal's own §2 measures these operators as correlated distance detectors, not independent evidence

Theoretical docking viability (fpocket):

pocket 1: score=0.305, druggability\_score=0.161  
pocket 2: score=0.112, druggability\_score=0.023  
pocket 3: score=0.091, druggability\_score=0.007  
pocket 4: score=0.081, druggability\_score=0.158  
pocket 5: score=0.054, druggability\_score=0.0